Describing Nuclear Quantum Effects in Coupled Nuclear-Electron Dynamics at Gas-Metal Surface

Rui-Hao Bi, Wenjie Dou

Laboratory of Theoretical and Computational Chemistry, Westlake University, Hangzhou, Zhejiang, China

Email: douwenjie@westlake.edu.cn

ISTCP 2024

西湖大學 WESTLAKE UNIVERSITY

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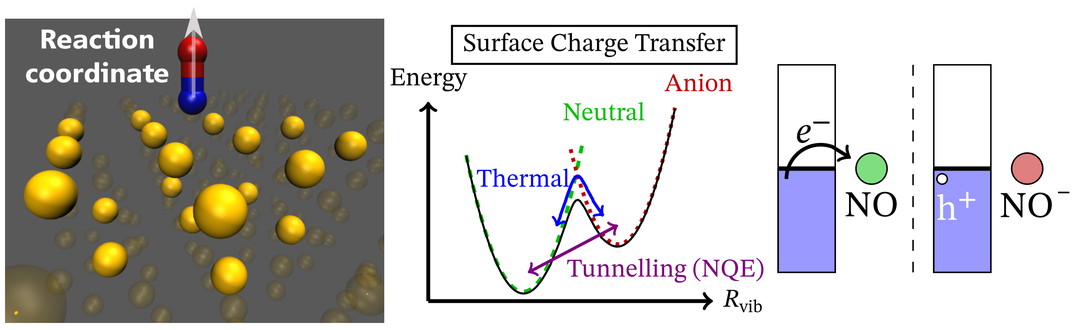
Problem Statement and Contribution

Goal: *Efficiently* incorporate nuclear quantum effects (NQE), such as tunnelling and zero-point energy, in coupled nuclear-electron dynamics at gas-metal interfaces.



Motivation:

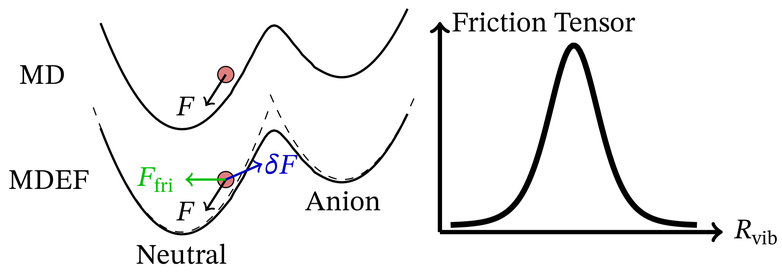
- Nonadiabatic dynamics at metal surfaces, including transient **charge transfer** and **energy dissipation**, are crucial for understanding surface species behavior.
- Mixed quantum-classical methods, such as Independent Electron Surface Hopping (IESH), Electronic Friction (EF), and Broadened classical Master Equation (BCME), all **lack** a consistent treatment of NQE, ignoring the quantum nature of molecular vibrations.

Key Contribution:

- A simple extension of the EF model to include NQE, which is **computationally efficient** and **easy to implement** in existing EF codes.
- Incorporation of NQE results in **improved** description of charge transfer dynamics as well as equilibrium distributions, **quantitatively matching** the results of full quantum dynamics from Quantum Master Equation (QME) simulations.

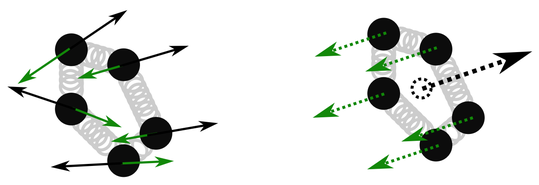
Theory

Main idea:



- The EF model is noting but ground-state Molecular Dynamics (MD) with the electronic friction and random forces; In MD, NQEs can be incorporated with **Ring Polymer Molecular Dynamics (RPMD)**, making it a natural choice.
- Yet, a naive implementation of RPMD leads to **cancellation of friction within a Ring Polymer**. To solve this, we propose a **centroid-mode electronic friction approximation**.

Naive Electronic Friction | **Cetroid Electronic Friction**



$F_{\rm fri}$: Local electronic friction — P: Bead mometa

$F^{\rm c}_{\rm fri}$: Centroid mode electronic friction — P: Centroid mode Mometa

Why Collective?

$$\sigma(R) \equiv \langle \hat{d}^\dagger \hat{d} \rangle,$$

$$\sigma^{\rm RP}_{\rm ss}(R) = \frac{1}{N}\sum_{i=1}^{N} \sigma^{(i)}_{\rm ss}(R_i),$$

$$\sigma^{(i)}(R) = \sigma^{(i)}_{\rm ss}(R) - \frac{1}{\Gamma}\frac{\partial \sigma^{\rm RP}_{\rm ss}}{\partial R_i}\frac{\partial R_i}{\partial R^{\rm c}}\frac{\partial R^{\rm c}}{\partial t}\delta t + \delta\sigma^{(i)}(R)$$

$$= \underbrace{\sigma^{(i)}_{\rm ss}(R)}_{\text{Mean potential}} - \underbrace{\frac{1}{\Gamma}\frac{\partial \sigma_{\rm ss}}{\partial R_i}\frac{P^{\rm c}}{M}\delta t}_{\text{Centroid friction}} + \underbrace{\delta\sigma^{(i)}(R)}_{\text{Random force}}$$

Model:

$$H = U_0(R) + E_d(R)\hat{d}^\dagger\hat{d} + \sum_k \epsilon_k \hat{c}^\dagger_k \hat{c}_k + \sum_k (V_k \hat{d}^\dagger \hat{c}_k + \text{h.c.}), \quad (1)$$

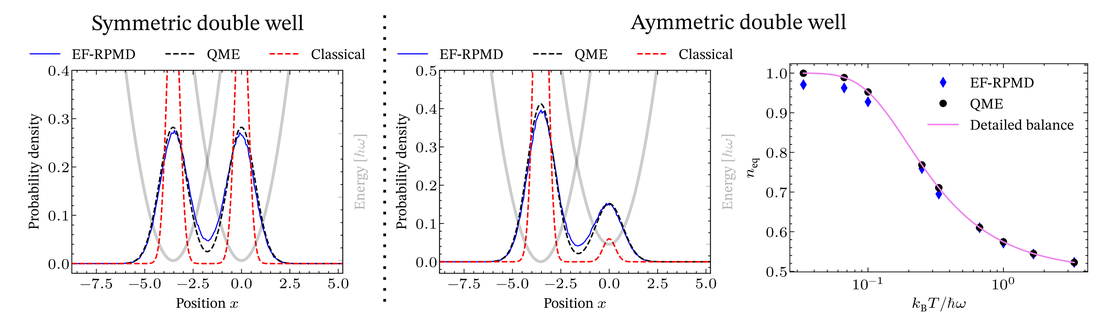
$$\hbar\Gamma = \hbar\Gamma(\epsilon) \equiv 2\pi \sum_k |V_k|^2 \delta(\epsilon_k - \epsilon) \quad (2)$$

Acknowledgements

This work was supported by the National Natural Science Foundation of China (NSFC) under Grant No. 22361142829.

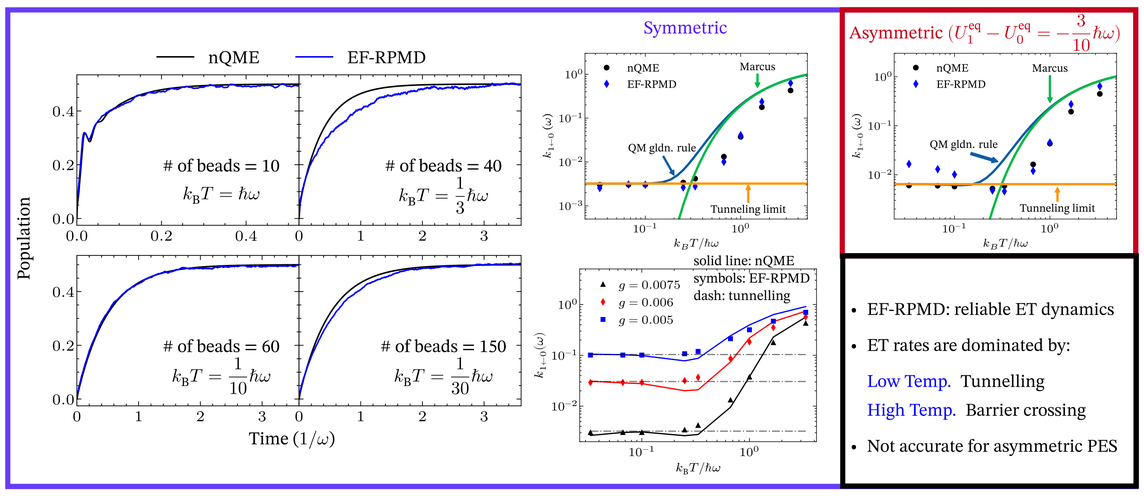
Results and Discussion

Equilibrium of Nuclear distribution and Electronic Population:



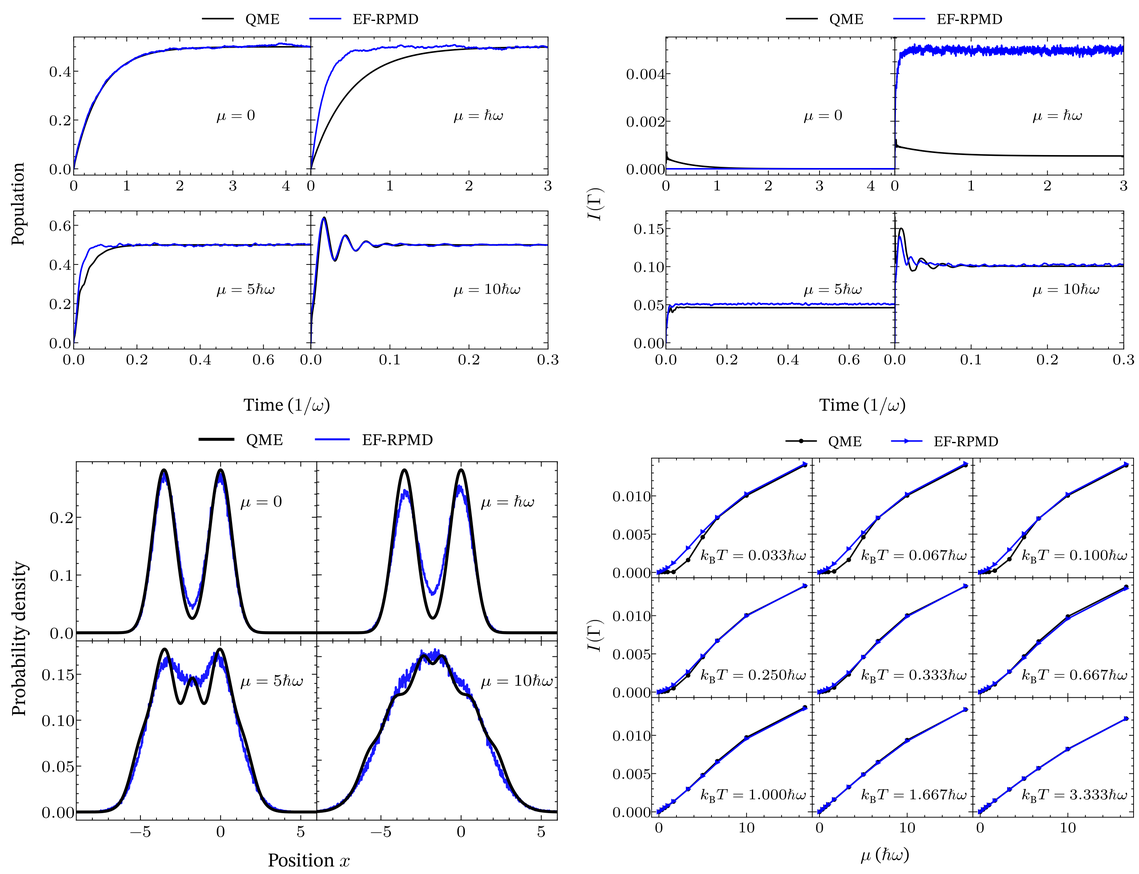
($k_{\rm B}T = \hbar\omega/10$ for nuclear distribution; Asymmetry: $U_0^{\rm eq} - U_1^{\rm eq} = \frac{3}{10}\hbar\omega$)

Charge transfer dynamics – thermal vs. tunnelling:



- EF-RPMD: reliable ET dynamics
- ET rates are dominated by:
  - Low Temp. Tunnelling
  - High Temp. Barrier crossing
- Not accurate for asymmetric PES

Non-equilibrium transport – dynamics and steady state:



Qualitatively correct (i) ET/current dynamics, (ii) steady-state distribution, and (iii) I-V curve, where notable deviations are present in small chemical potential (μ) bias.

Perspective

In this work, we efficiently incorporated NQEs in EF. Yet, many subjects remain unexplored:

- **Limitations of the EF:** EF fails when Γ is small (strongly nonadiabatic). By considering the memory effects of EF, can we further improve the current model?
- **Vibrational energy relaxation,** a focus in gas-metal scattering experiments, is not considerred in this work due to the thermal equilibrium nature of RPMD. How will NQEs affect the energy dissipation (non-equilibrium), and how to describe it?

Paper and Code

Code

J. Chem. Phys. 160, 074110 (**2024**)